

Mahidol University
Faculty of Information
and Communication Technology

Graphical Interface, System Configuration, and Common Applications

ITCS 254 | Lecture 2

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Agenda

- Graphical desktop
- Session management
- Basic operations
- System configuration
- Common applications

Learning Objectives



Manage graphical interface sessions.



Perform basic operations using the graphical interface.



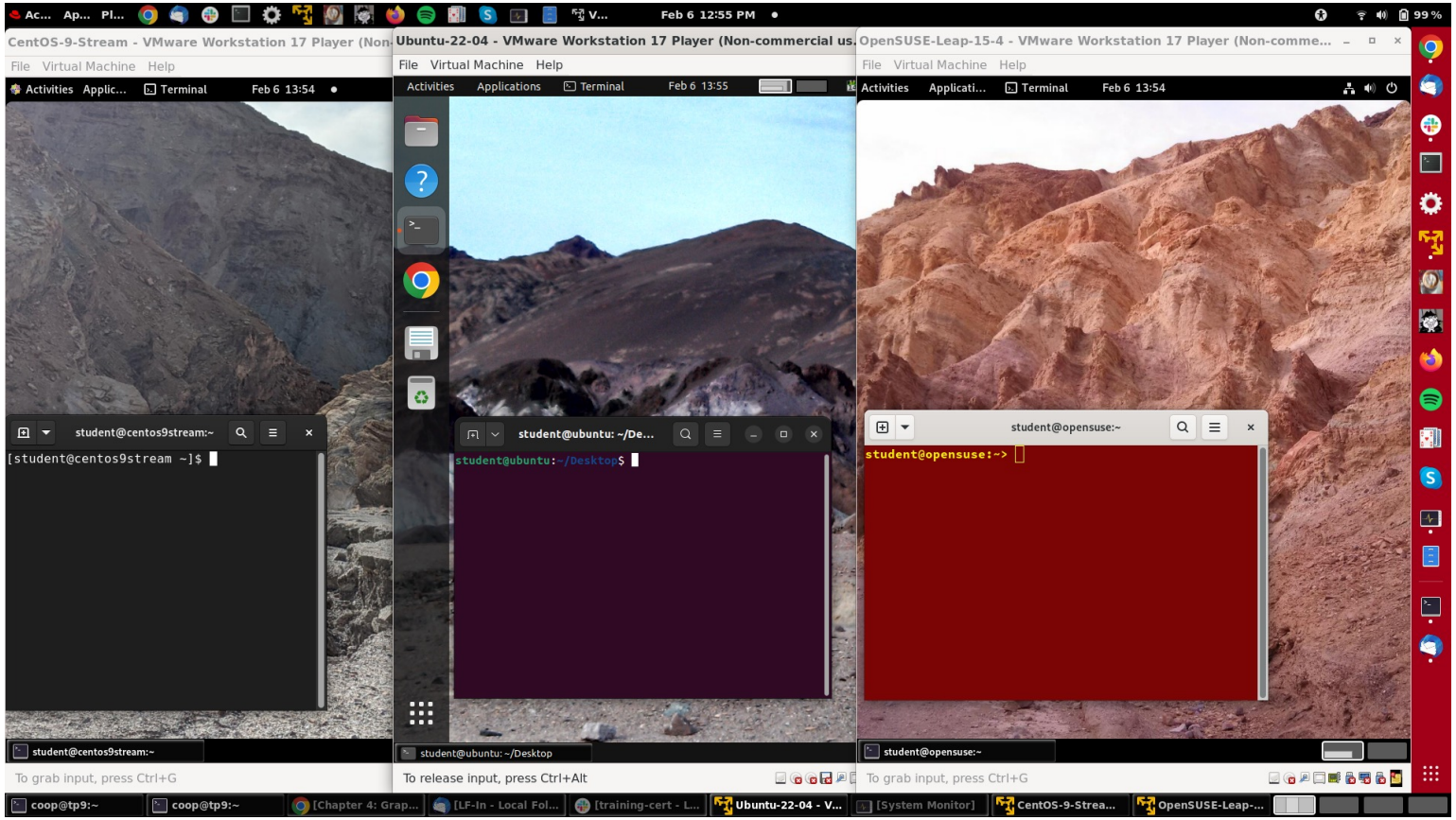
Change the graphical desktop to suit your needs.



Know common open source applications on Linux

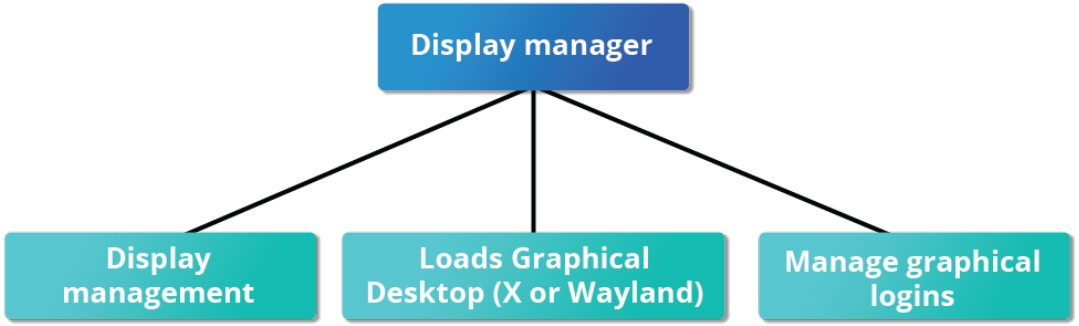
Graphical Desktop

CentOS, Ubuntu, and openSUSE Desktops



X Window System

- Loading the graphical desktop is one of the final steps in the boot process of a Linux desktop.
- A service called the Display Manager keeps track of the displays being provided and loads the X server.
  - So-called, because it provides graphical services to applications, sometimes called X clients.
  - It also handles graphical logins and starts the appropriate desktop environment after a user logs in.
- X is rather old software; it dates back to the mid-1980s and, as such, has certain deficiencies on modern systems (such as, with security), as it has been stretched rather far from its original purposes.
- A newer system, known as Wayland, is gradually superseding it and is the default display system for Fedora, RHEL, and other recent distributions.
  - For the most parts, it looks just like X to the user, although under the hood it is quite different.



GUI Startup

- When you install a desktop environment, the display manager starts at the end of the boot process.
- It is responsible for starting the graphics system, logging in the user, and starting the user's desktop environment.
- You can often select from a choice of desktop environments when logging in to the system.
- The default display manager for GNOME is called gdm. Another popular display manager is kdm, associated with KDE.

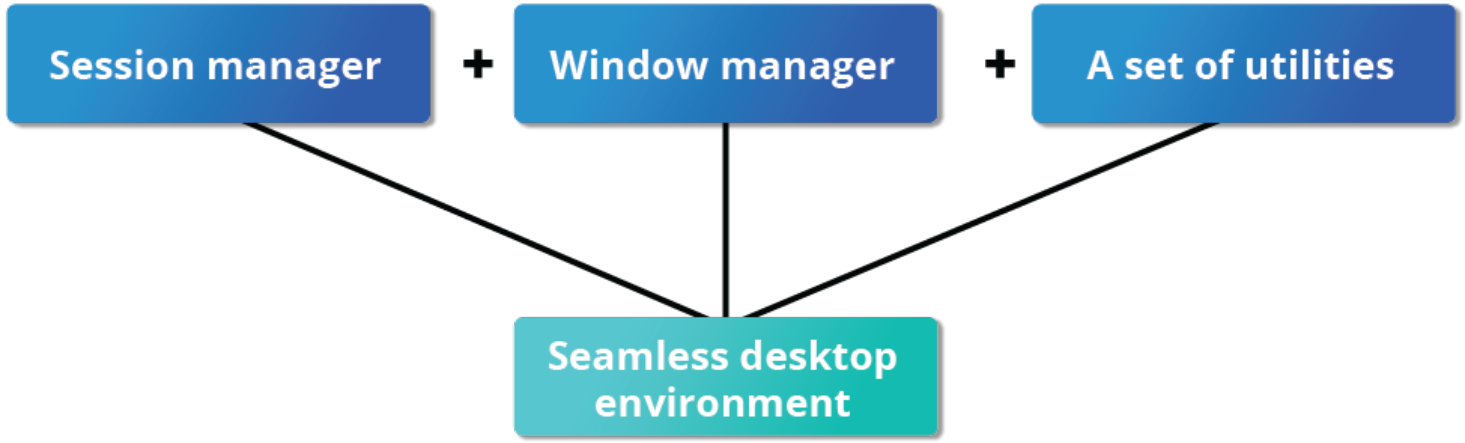
Popular Desktop Environments

- **GNOME** is a popular desktop environment with an easy-to-use graphical user interface.
  - It is bundled as the default desktop environment for most Linux distributions.
  - Used in Red Hat Enterprise Linux (RHEL), Fedora, CentOS, SUSE Linux Enterprise, Ubuntu, and Debian
  - GNOME has menu-based navigation and is sometimes an easy transition to accomplish for Windows users.
  - The look and feel can be quite different across distributions, even if they are all using GNOME.
- **KDE** has often been used in conjunction with SUSE and openSUSE.
- Other alternatives include Unity (present on older Ubuntu but still based on GNOME), XFCE, and LXDE.
- Most desktop environments follow a similar structure to GNOME.





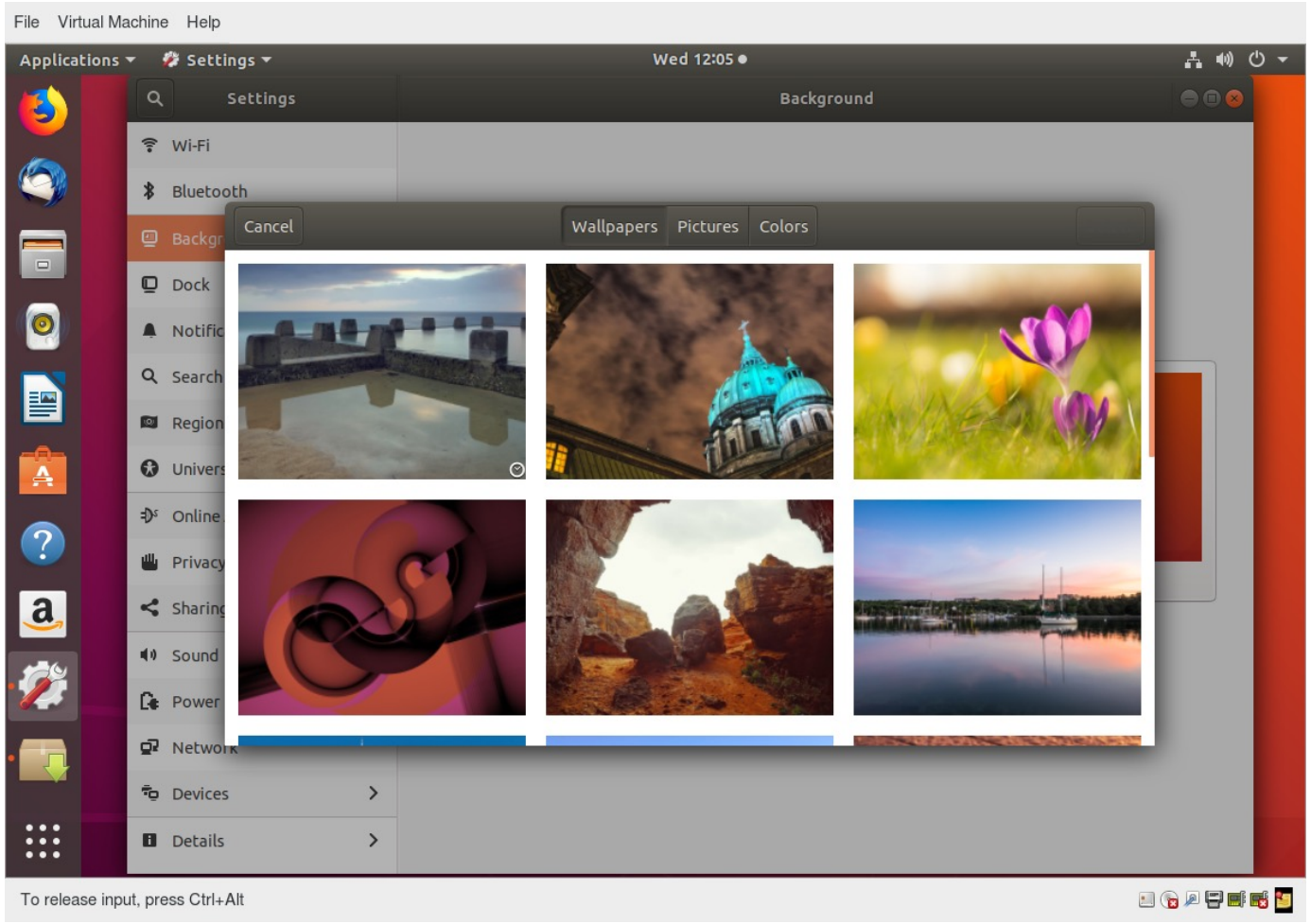
Desktop Environment



- If the display manager is not started by default in the default runlevel, you can start the graphical desktop different way, after logging on to a text-mode console, by running `startx` from the command line.
    - Or, you can start the display manager (`gdm`, `kdm`, `xdm`, etc.) manually from the command line. This differs from running `startx` as the display managers will project a sign in screen.

Customizing the Desktop Background

Try-it-yourself: [Changing the Desktop Background in openSUSE]



Session Management [1/2]

- Logging in and out using GUI
- Locking and unlocking the screen. There are two ways:
  - Using the graphical interface: Clicking in the upper-right corner of the desktop, and then clicking on the lock icon.
  - Using the keyboard shortcut SUPER-L (or SUPER-Escape) (The SUPER key is also known as the Windows key).
- Switching users
  - Switching users vs logging out
  - Try: http://linuxfoundation.s3-website-us-east-1.amazonaws.com/TIY/switchuserubuntu/index.html
- Shutting down and restarting

Session Management [2/2]

- Suspending or sleep mode
  - When you want to stop using your computer for a while. Suspend Mode saves the current system state and allows you to resume your session more quickly while remaining on but uses very little power in the sleeping state.
  - It works by keeping your system's applications, desktop, and so on, in system RAM, but turning off all of the other hardware.
  - This shortens the time for a full system start-up as well as conserves battery power.

Basic operations

Locating applications

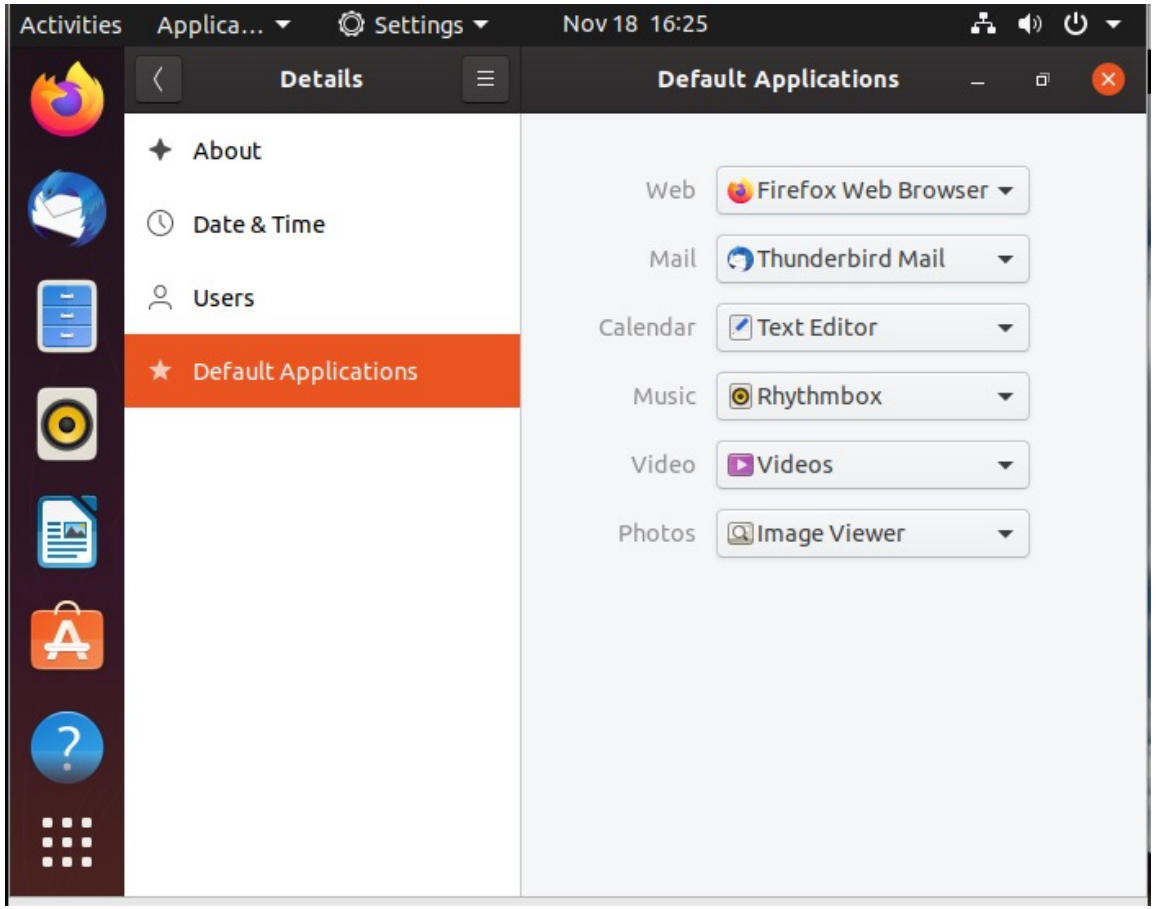
Default applications

File manager

- Home directory
- Viewing files
- Searching for files
- Editing a file
- Removing a file

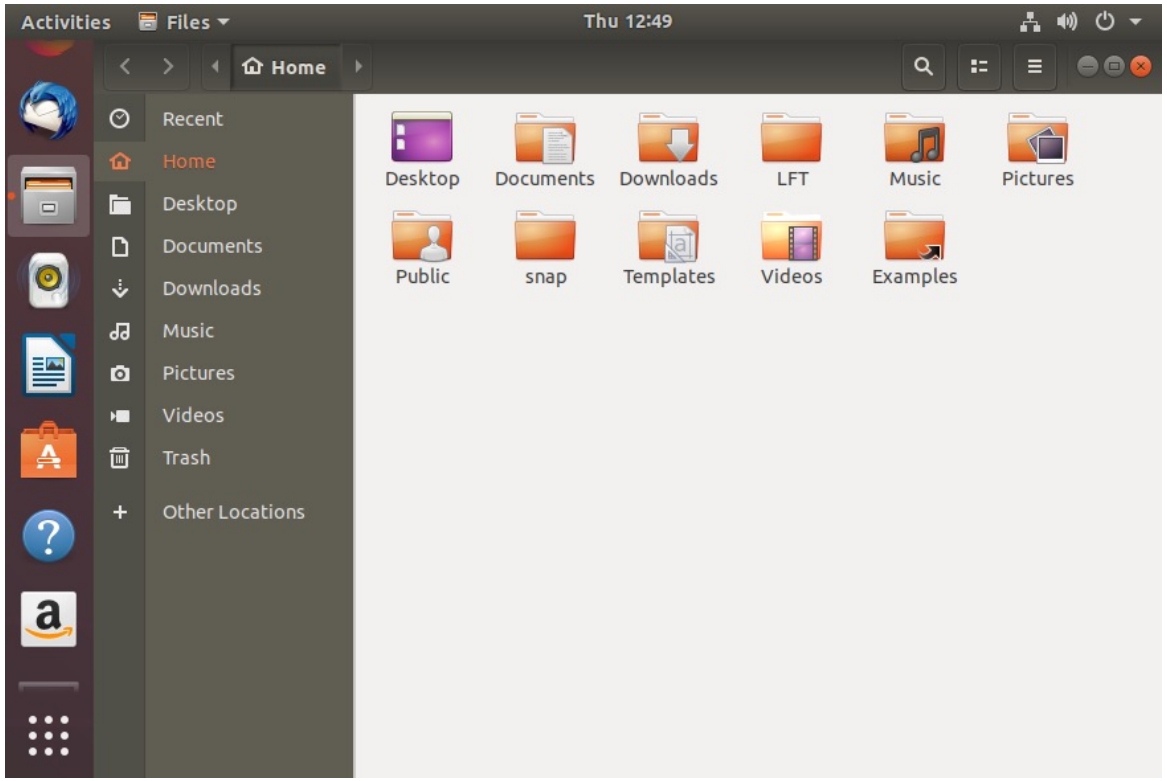
Default Applications

- Setting > Default Applications or Details > Default Applications.
- On Ubuntu: Setting > Apps > Default Apps
- The exact list will vary according to what is actually installed and available on your system.



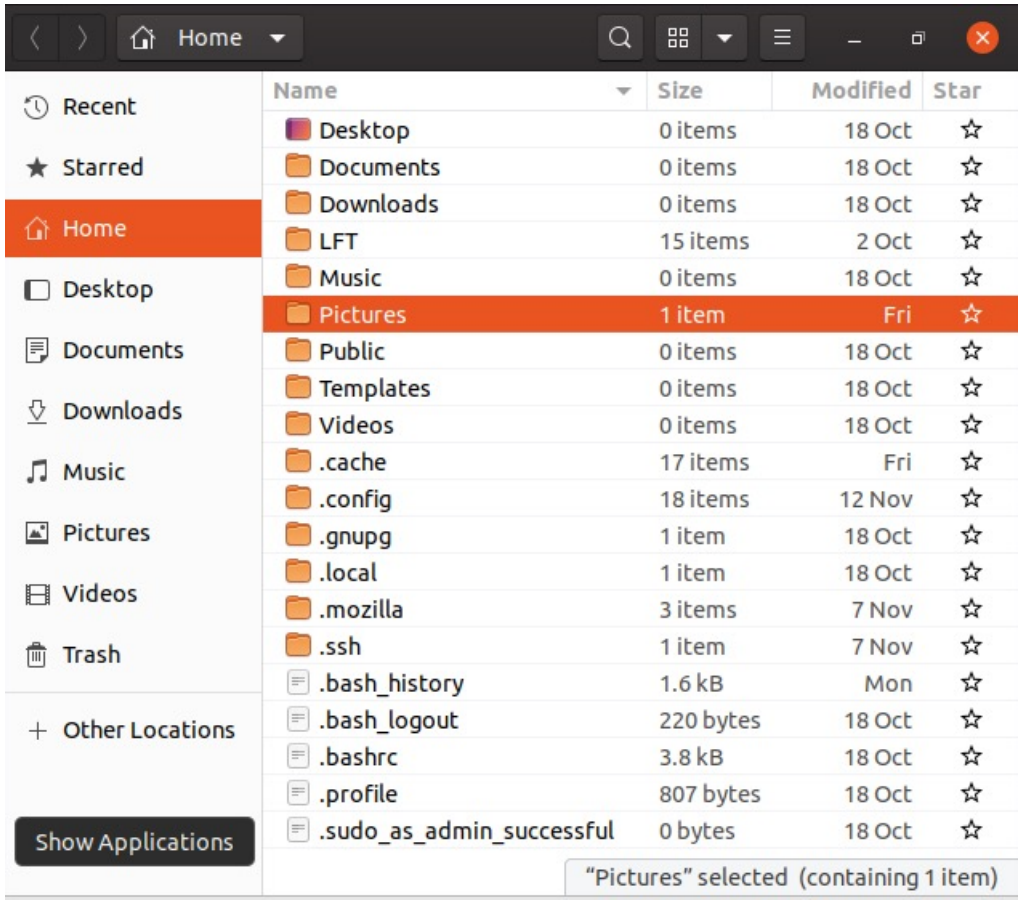
File Manager

- Each distribution implements the Nautilus (File Manager) utility
- Command line to load file manager: `nautilus`
- You can click the magnifying glass icon on the top-right to search for files or directories (folders).



Home Directory

- Every user with an account on the system will have a home directory, usually created under `/home`, and usually named according to the user, such as `/home/student`.
- By default, files the user saves will be placed in a directory tree starting there.

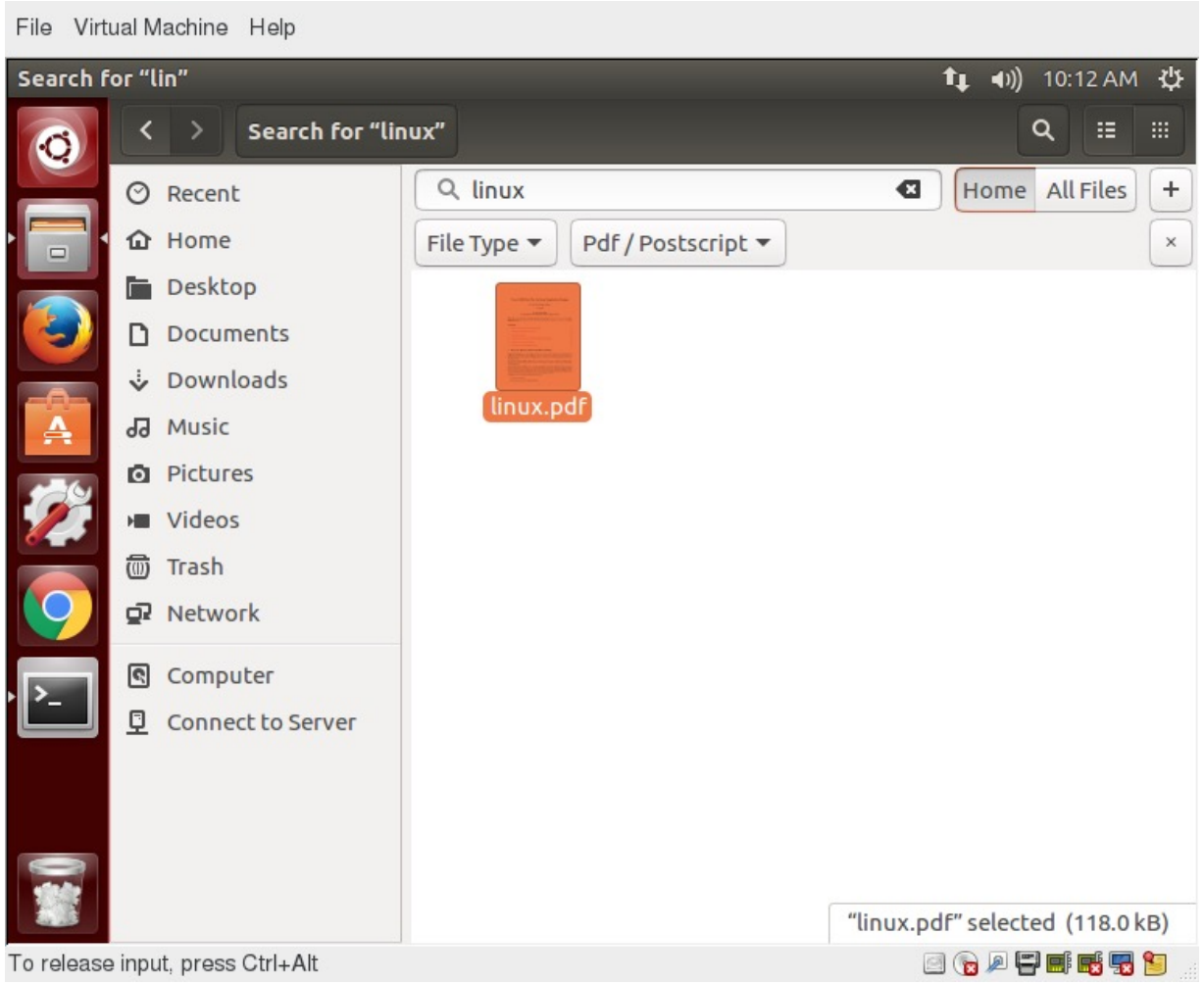


Viewing a File

- You can switch between the Icons and List formats, either by clicking the familiar icons in the top bar, or you can press CTRL-1 or CTRL-2 respectively.
- You can also arrange the files and directories by name, size, type, or modification date for further sorting.
- Another useful option is to show hidden files (sometimes imprecisely called system files), which are usually configuration files that are hidden by default and whose name starts with a dot.
    - To show hidden files, select Show Hidden Files from the menu or press CTRL-H.

Searching for Files

- The shortcut key to get to the search text box is CTRL-F.
- Another quick way to access a specific directory is to press CTRL-L, which will give you a Location text box to type in a path to a directory.



Editing a File

- The default text editor in GNOME is `gedit`.
- It is simple yet powerful, ideal for editing documents, making quick notes, and programming.
- Although `gedit` is designed as a general-purpose text editor, it offers additional features for spell-checking, highlighting, file listings, and statistics.
- On Ubuntu Desktop, we have Text Editor app.

Removing a File

- Deleting a file or directory using Nautilus
  - Select all the files and directories that you want to delete.
  - Press CTRL-Delete on your keyboard, or right-click the file then select Move to Trash.
- Deleting a file in Nautilus will automatically move the deleted files to the `.local/share/Trash/files/` directory (a trash can of sorts) under the user's home directory.
- Note that you may have a Delete Permanently option which bypasses the trash folder, and that this option may be visible all the time or only in list (rather than) icon mode.
- To permanently delete a file
  - On the left panel inside a Nautilus file browser window, right-click on the Trash directory and Select Empty Trash.
  - Alternatively, select the file or directory you want to permanently delete and press Shift-Delete.
- As a precaution, you should never delete your Home directory, as doing so will most likely erase all your GNOME configuration files and possibly prevent you from logging in.

System Configuration



System, display, and date/time settings

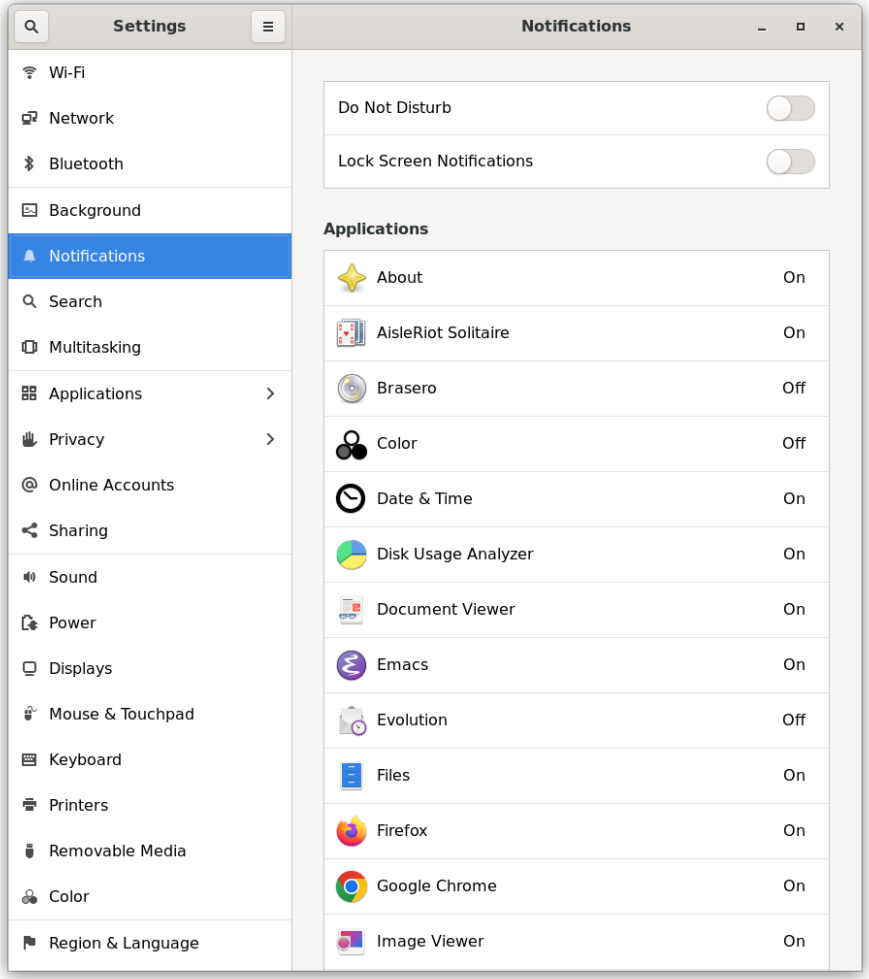


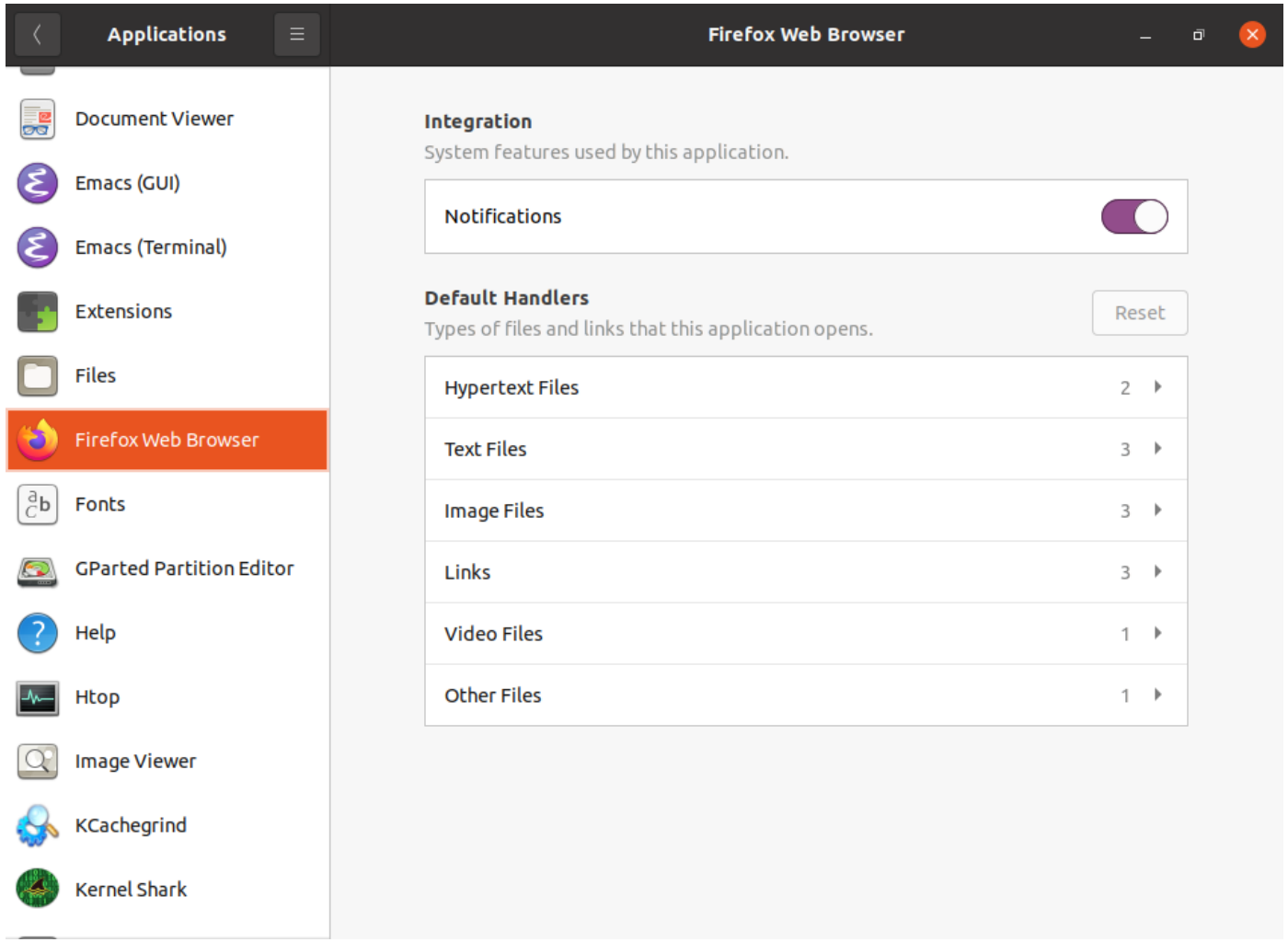
Network manager



Installing and updating software

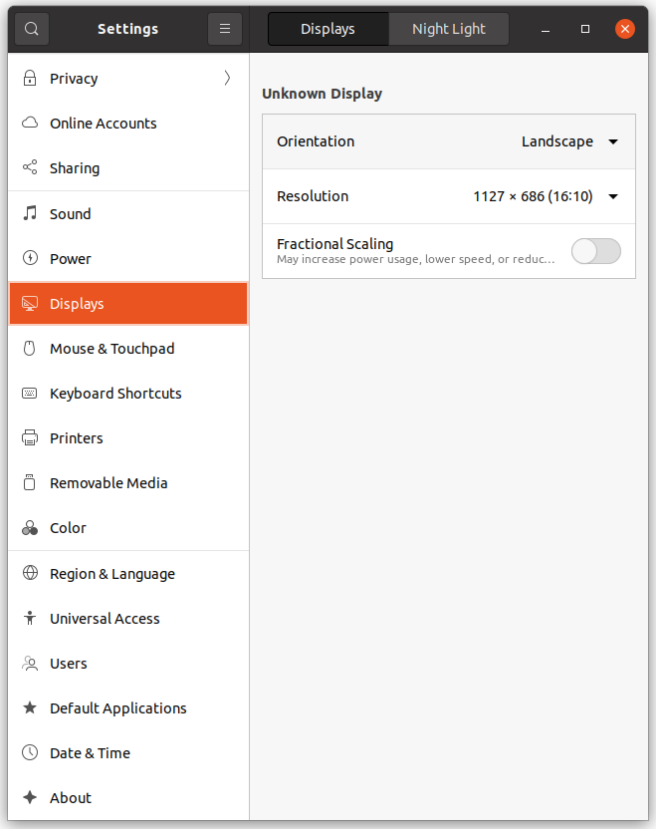
System Setting

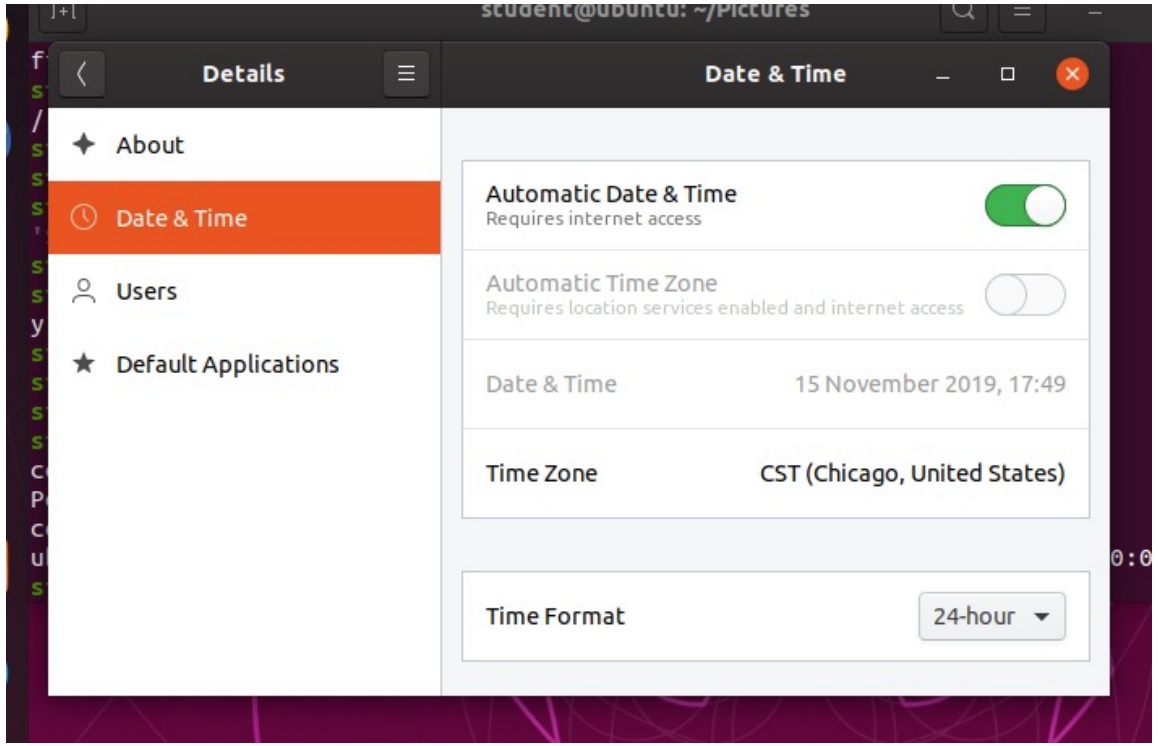




Display and Date/Time Setting

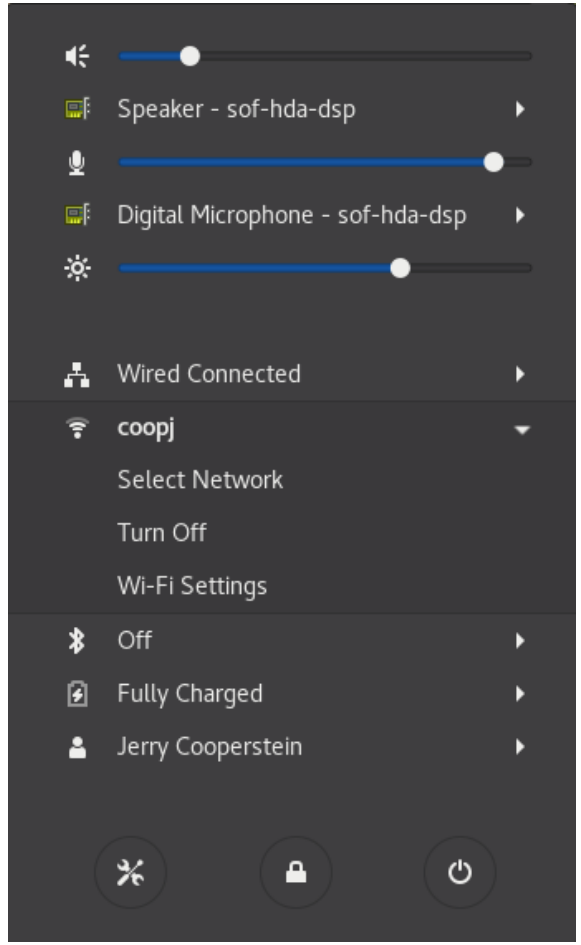
Settings > Displays (or Settings > Devices > Displays)



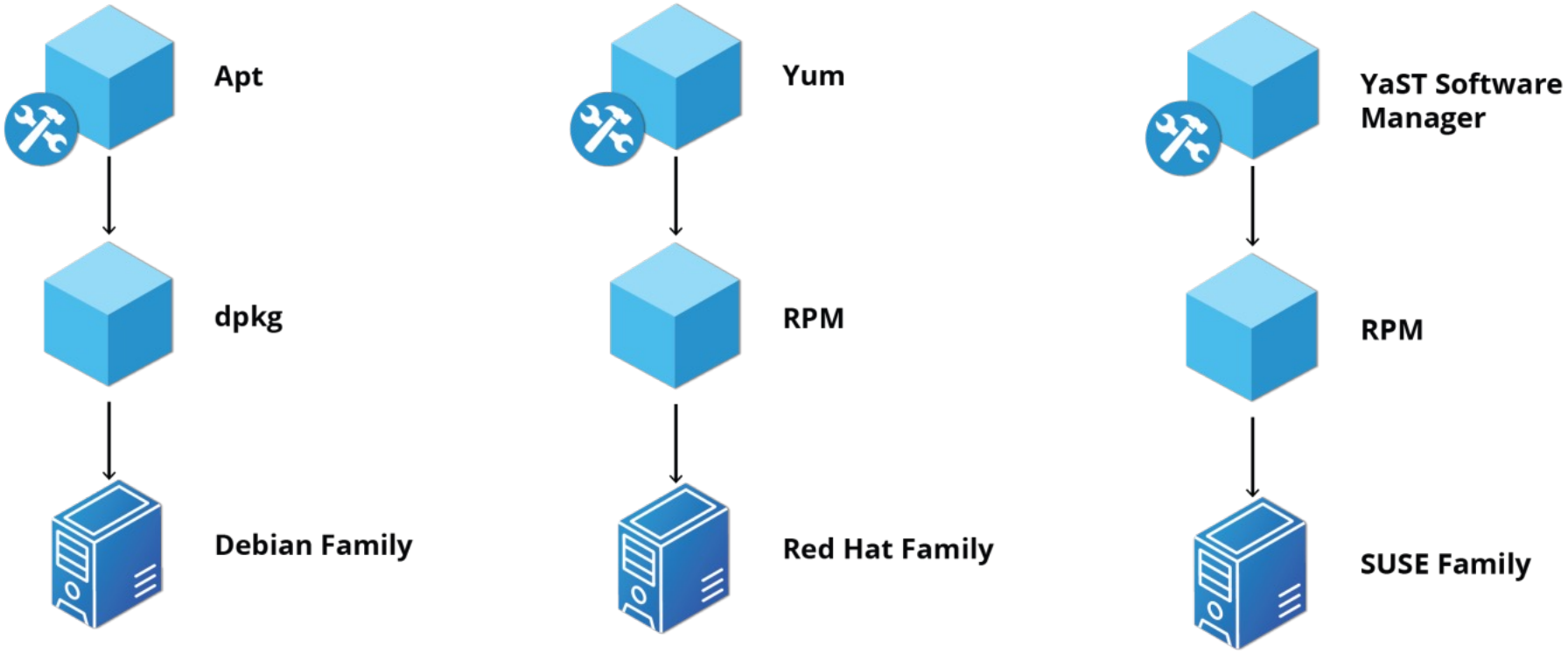


Network Manager

- Wired connections usually do not require complicated or manual configuration.
- The hardware interface and signal presence are automatically detected, and then Network Manager sets the actual network settings via Dynamic Host Configuration Protocol (DHCP).
- For static configurations that do not use DHCP, manual setup can also be done easily through Network Manager.



Package Management



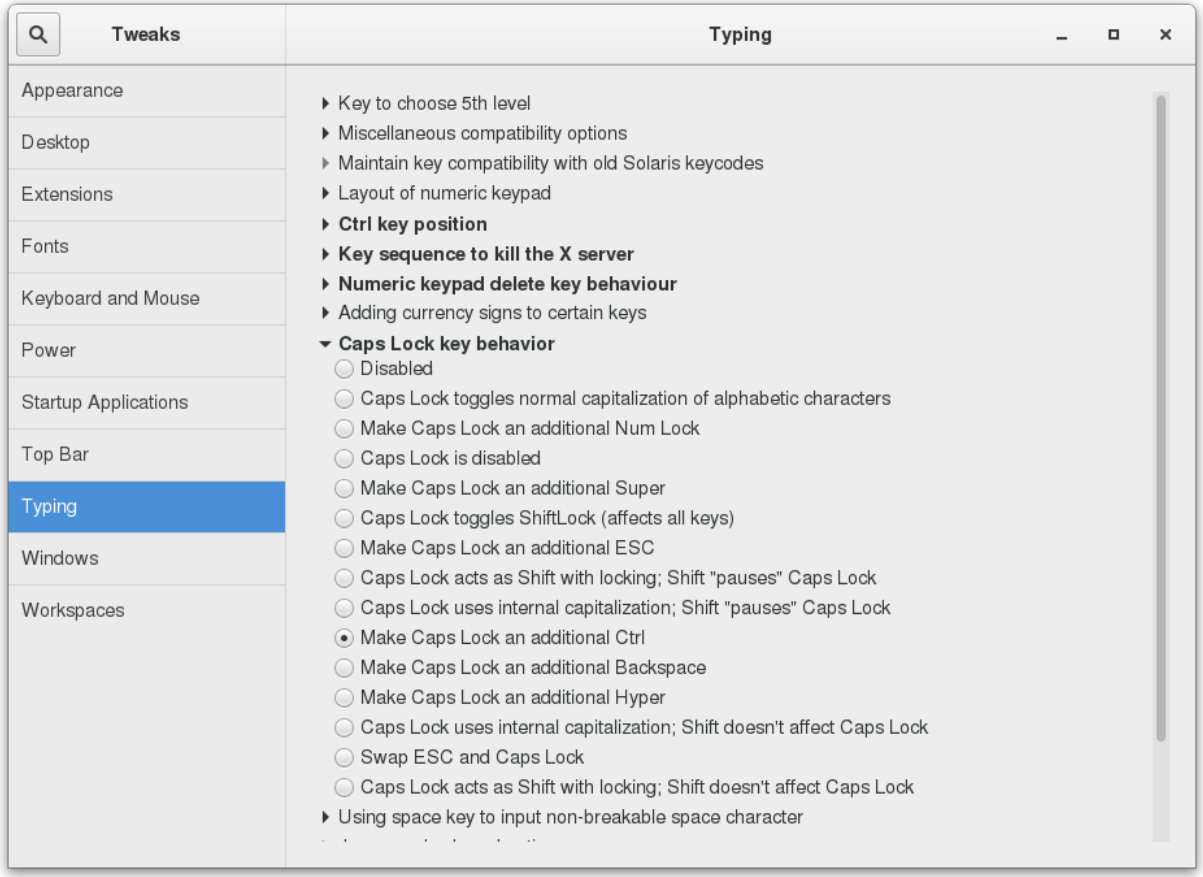
Installing and Removing Software Package

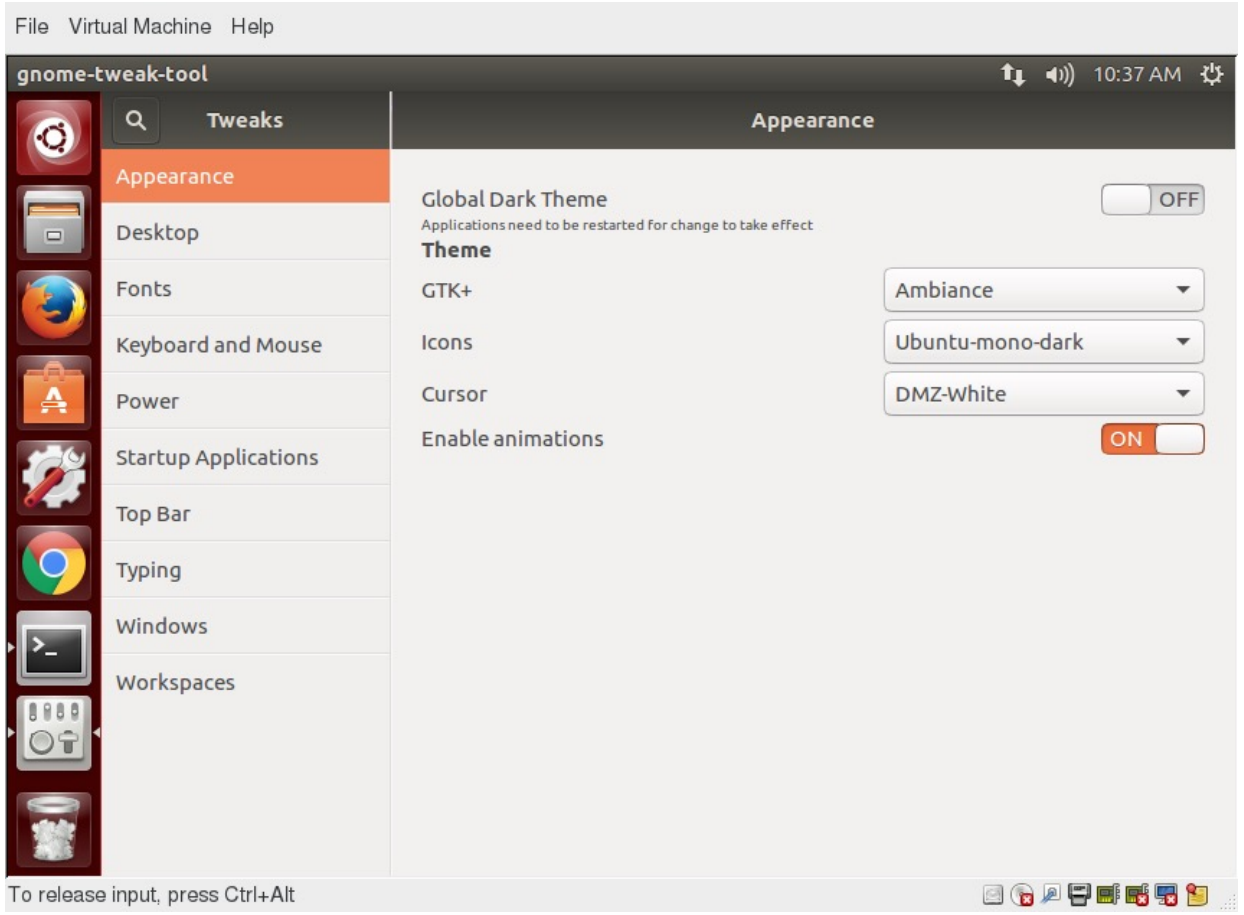
- On any recent GNOME-based system, you can run the gnome-software utility as we have described. You can either run it from some menu location, or from a command line, or by typing it after hitting ALT-F2.
- On Ubuntu, use either the Ubuntu Software Center or synaptic. On any other Debian-based system, you can use synaptic.
- On Red Hat 7-based systems, including CentOS 7, click on Applications in the top menu bar, then System Tools > Software.
- On SUSE-based systems, you can click on the System Settings icon in the upper right corner (the wrench and screwdriver) and then System > YaST.

gnome-tweaks

- There are many settings which many users would like to modify which are not thereby accessible; the default settings utility is unfortunately rather limited in modern GNOME-based distributions.
- gnome-tweaks is a standard utility, which exposes many more setting options.
- It also permits you to easily install extensions by external parties.
- Not all Linux distributions install this tool by default, but it is always available (older distributions used the name gnome-tweak-tool).
- You may have to run it by hitting Alt-F2 and then typing in the name.
- Some recent distributions have taken most of the functionality out of this tool and placed it in a new one called gnome-extensions-app.
- Note that, on Ubuntu Desktop, we need to run `gnome-tweaks` in Terminal.

gnome-tweaks Examples





Common Applications



Internet applications



Productivity and development applications

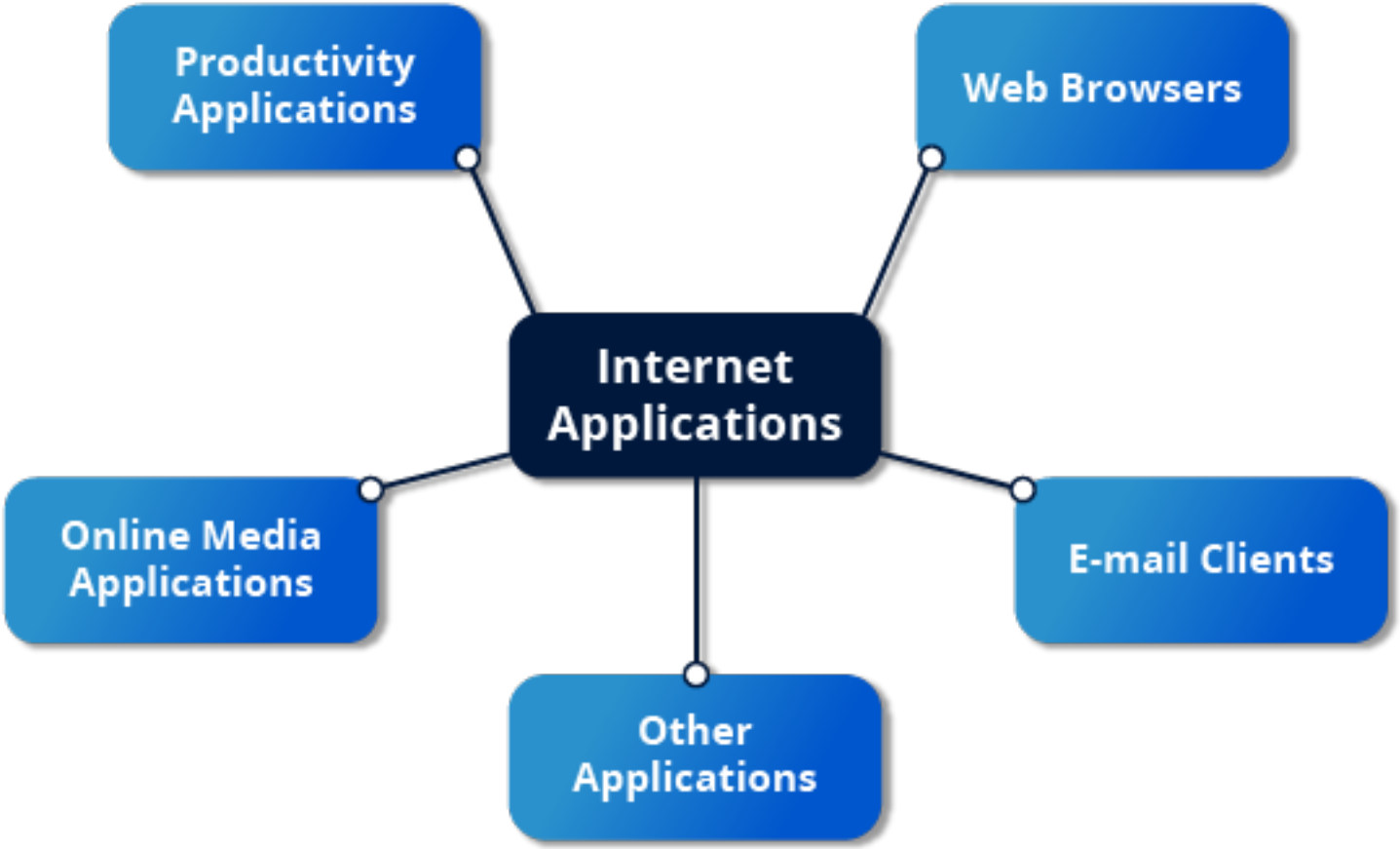


Multimedia applications



Graphics editors and utilities

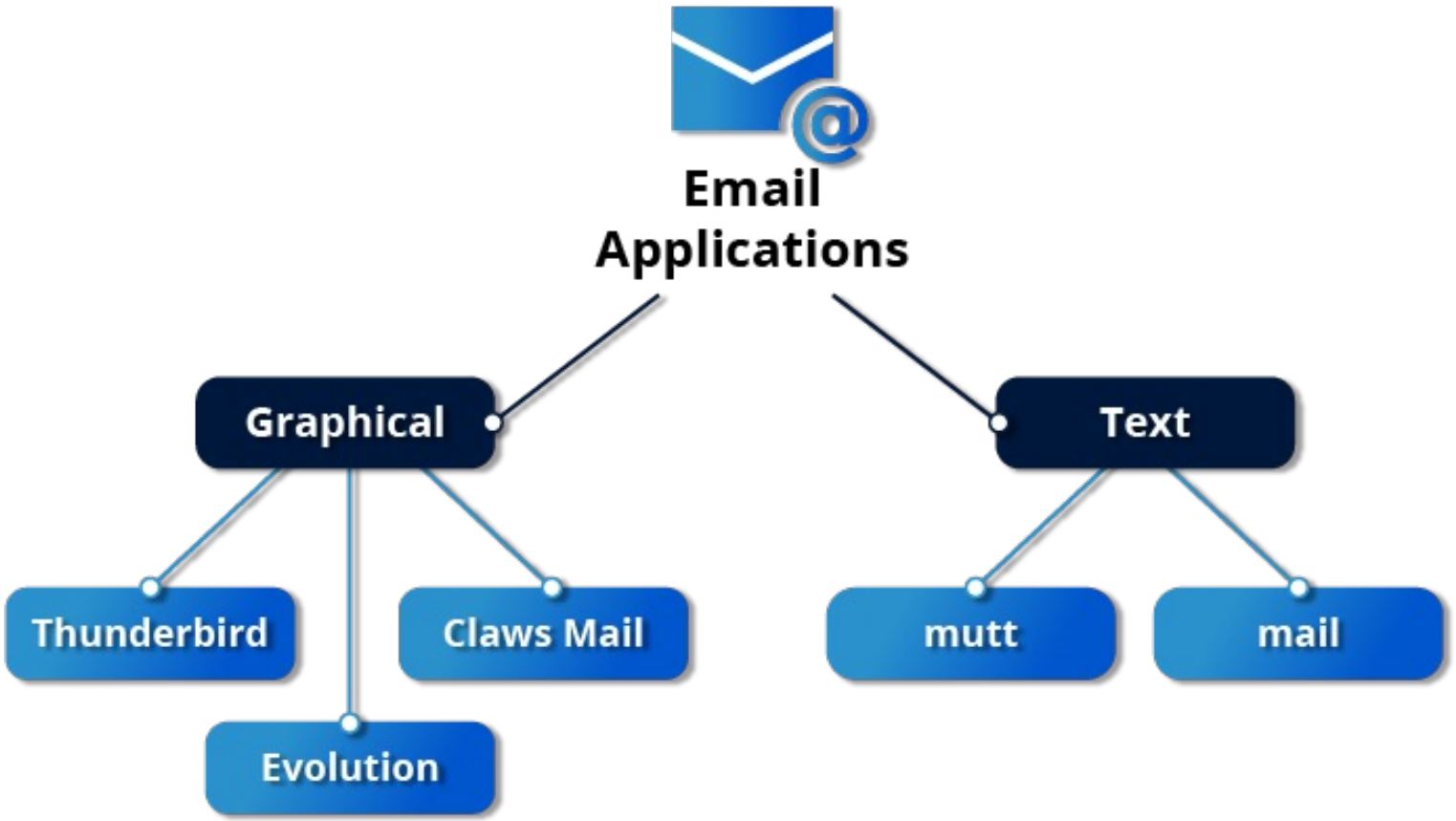
Internet Applications



Web Browsers

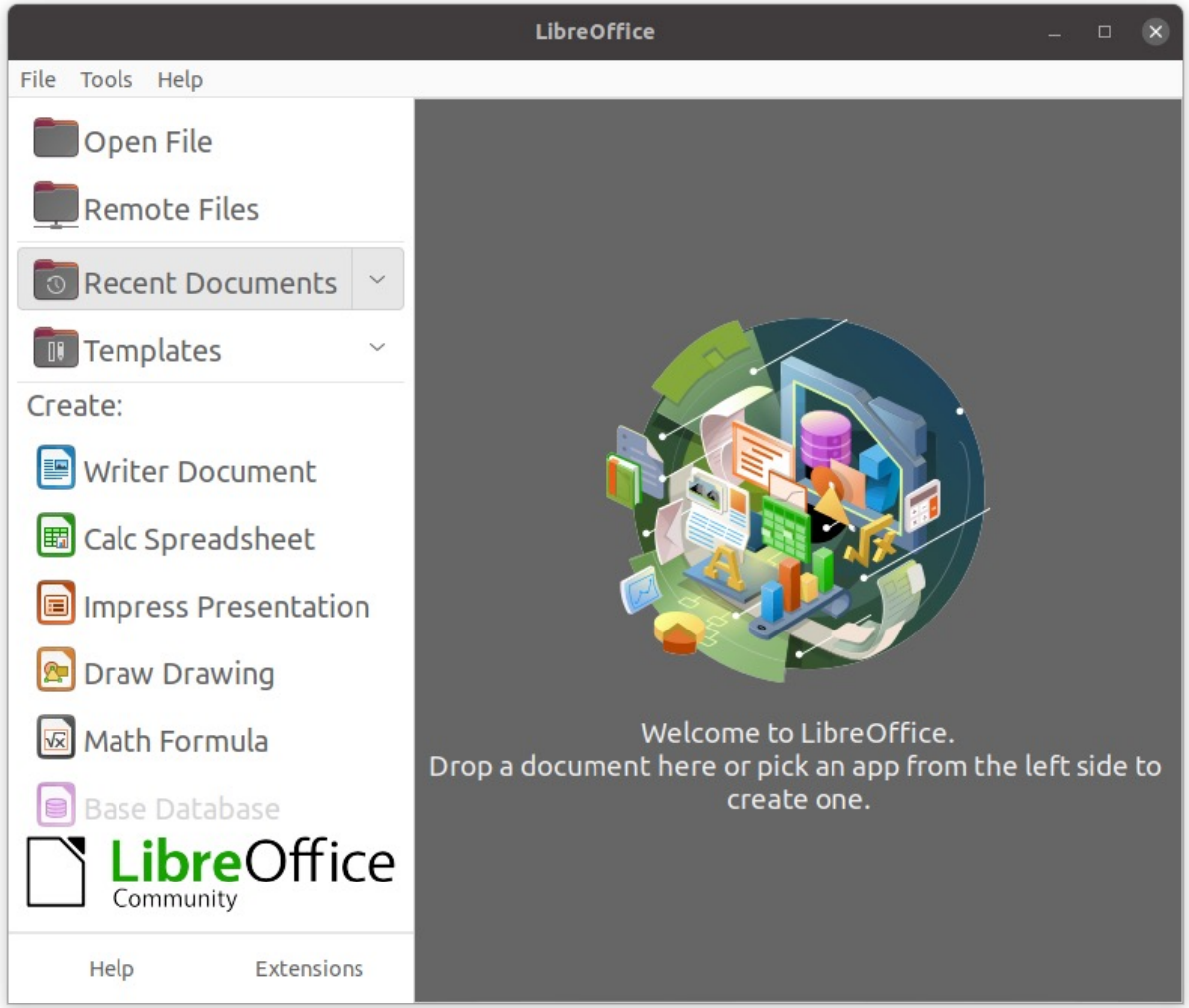
- Firefox
- Google Chrome
- Chromium
- Epiphany (renamed web)
- Konqueror
- Iinx, lynx, w3m
- Opera

Email Applications



Office Applications: LibreOffice

- Most Linux distributions offer LibreOffice, an open source office suite that started in 2010 and has evolved from OpenOffice.
- While other office suites are available, LibreOffice is the most mature, widely used and intensely developed.



Development Applications

- Advanced editors customized for programmers' needs, such as `vi` and `emacs`.
- Compilers (such as `gcc` and `clang` for programs in C and C++) for every computer language that has ever existed, including very popular new ones such as Golang and Rust.
- Debuggers such as `gdb` and various graphical front ends to it, and many other debugging tools (such as Valgrind).
- Performance measuring and monitoring programs, some with easy to use graphical interfaces, others more arcane and meant to be used only by serious experienced development engineers.
- Source and Revision control systems, such as git (and associated interfaces to it, including GitHub and GitLab) and older systems, such as Apache Subversion.
- Complete Integrated Development Environments (IDE's) such as Eclipse and Visual Studio Code that put all these tools together.

Sound Players



Amarok

- Mature MP3 player with a graphical interface, that plays audio and video files, and streams (online audio files).
- It allows you to create a playlist that contains a group of songs, and uses a database to store information about the music collection.



Audacity

- Used to record and edit sounds. Audacity has a simple interface to get you started.



Audacious

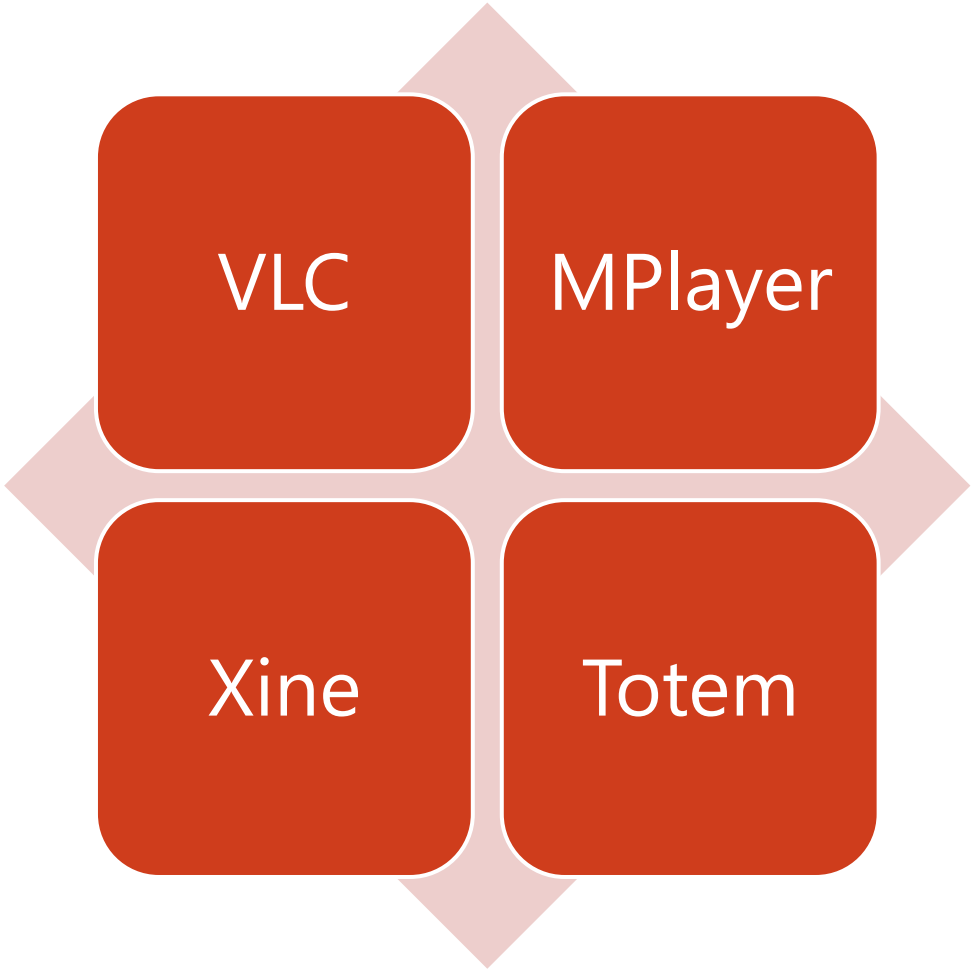
- Another smart audio media player.



Rhythmbox

- Supports a large variety of digital music sources, including streaming Internet audio and podcasts.
- The application also enables search of particular audio in a library.
- It supports smart playlists with an automatic update feature, which can revise playlists based on specified selection criteria.

Movie Players



Movie Editors



Blender

- Creates 3D animation and design.
- Blender is a professional tool that uses modeling as a starting point.
- There are complex and powerful tools for camera capture, recording, editing, enhancing and creating video, each having its own focus.



Cinelerra

- Capture, compose, and edit audio/video.

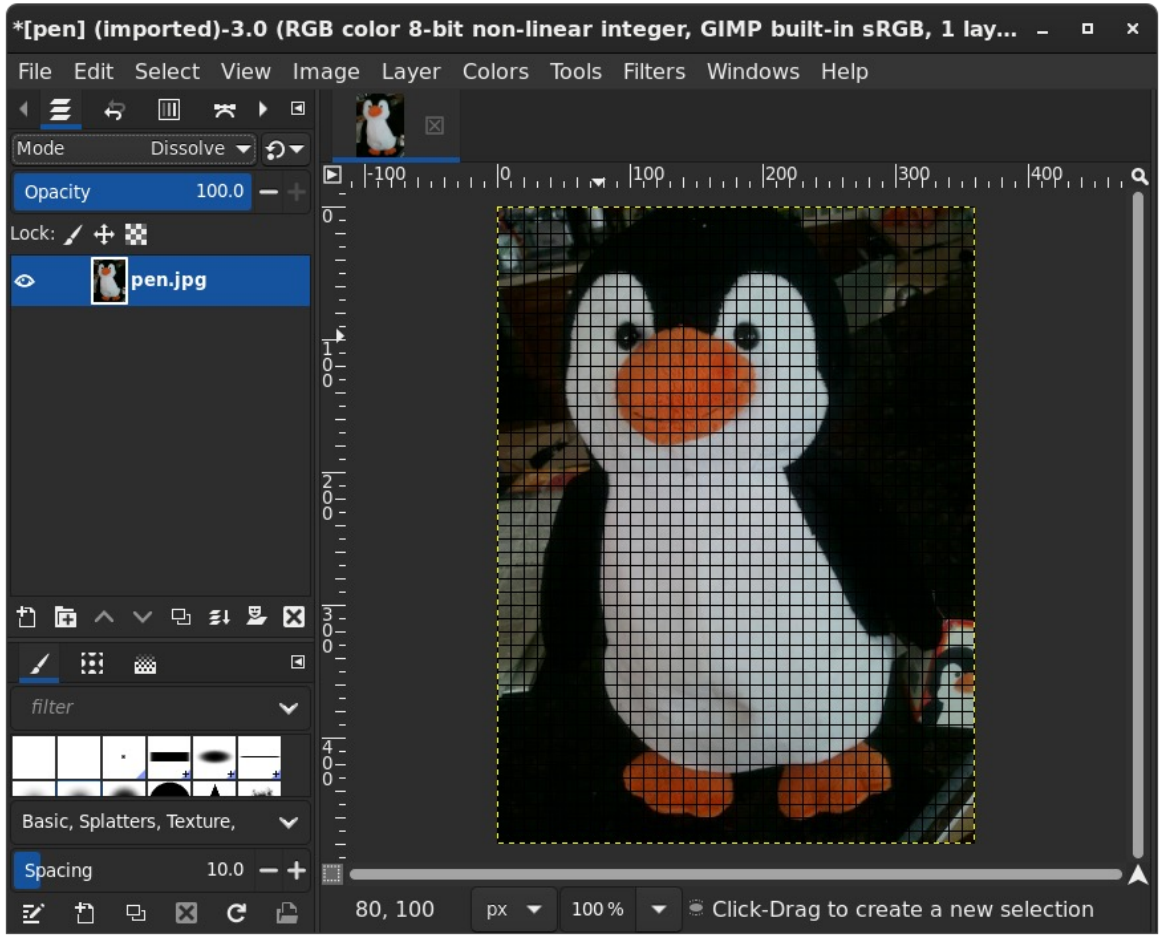


FFmpeg

- Record, convert, and stream audio/video. FFmpeg is a format converter, among other things, and has other tools such as ffplay and ffserver.

Graphics Editors

The GNU Image Manipulation Program (GIMP) is a feature-rich image retouching and editing tool similar to Adobe Photoshop and is available on all Linux distributions.



Graphic Utilities



Eye of Gnome (eog)

- It is an image viewer that provides slide show capability and a few image editing tools, such as rotate and resize. It can also step through the images in a directory with just a click.



Inkscape

- It is an image editor with lots of editing features. It works with layers and transformations of the image. It is sometimes compared to Adobe Illustrator.



convert

- It is a command line tool (part of the ImageMagick set of applications) that can modify image files in many ways. The options include file format conversion and numerous image modification options, such as blur, resize, despeckle, etc.



Scribus

- It is used for creating documents used for publishing and providing a *What You See Is What You Get (WYSIWYG)* environment. It also provides numerous editing tools.